Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

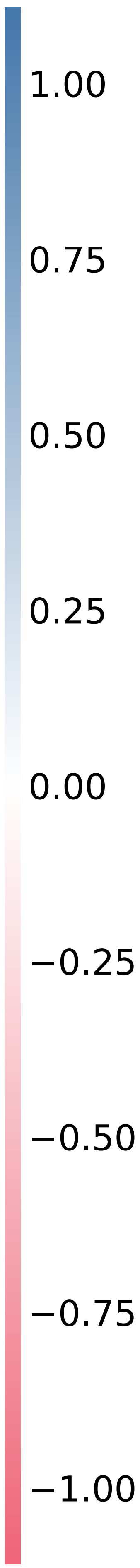
F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV



Reasoning + Answer:

<think>

reasoning process here

de Broglie wavelength is given by Î» = h/p, where h is Planck's constant and p is momentum, which is âĪļ(2*me*Ek), where me is the mass of the electron and Ek is the kinetic energy.

Planck's constant, h = 6.625 ÃĹ10^-27 erg sec.

The mass of the electron, me = 9.109 ÃĹ10^-28 g = 9.109 ÃĹ10^-31 kg.

Kinetic energy, Ek = 1 eV = 1.602 ÃĹ10^-19 J.

Therefore, momentum, p = âĪļ(2*me*Ek) = âĪļ(2*9.109 ÃĹ10^-31*1.602 ÃĹ10^-19) = 5.923 ÃĹ10^-26 kg m/s.

So, de Broglie wavelength, Î» = h/p = 6.625 ÃĹ10^-27 / 5.923 ÃĹ10^-26 = 1.127 ÃĹ10^-1 = 1.127 ÃĹ10^-1 cm.

By approximation, Î» = 1.127 ÃĹ10^-1 cm = 11.27 ÃĹ10^-2 cm ~ 10^-7 m.

</think>

<answer>

F. 5.27 ÃĹ10^-9 m

</answer>